

Task 16 — Hypothesis Testing & A/B Experimentation Report

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1. EXPERIMENT DESIGN & POWER CALCULATION

Hypothesis (H0): Mean performance of Control (Group A) and Variant (Group B) are equal ($\mu_A = \mu_B$).
Alternative (H1): Variant B produces higher performance ($\mu_B > \mu_A$).
Power Analysis: Prior to running the test, sample size calculation was fixed with significance level $\alpha = 0.05$ and target power $1 - \beta = 0.80$ for medium effect size (Cohen's $d = 0.40$). Required = 100 per group (total = 200).

2. EXPERIMENTAL RESULTS & METRIC BREAKDOWN

Metric / Statistic	Control (Group A)	Variant (Group B)	Difference / Test Stat	Verification Status
Sample Size (N)	100	100	200 Total Observations	Balanced Allocation
Mean Score (μ)	65.41	72.84	+7.43 Improvement	Observed Increase
Std Deviation (σ)	17.62	18.15	Pooled Std = 17.89	Homogeneity Checked
t-Statistic / p-Value	—	—	$t = 2.936, p = 0.0037$	$p < 0.05$ (Stat Sig)
Effect Size (Cohen's d)	—	—	$d = 0.415$	Medium Effect Size
95% Confidence Interval	—	—	[+2.44, +12.42]	Zero Excluded

3. PITFALL SAFEGUARDS & RECOMMENDATION

- Anti-P-Hacking Safeguard:** Sample size was pre-committed at 100/group before testing; no early stopping was performed.
- Effect Practical Significance:** The observed boost (0.43 points, Cohen's $d = 0.415$) confirms a practical, non-trivial impact beyond statistical significance.
- Recommendation:** Reject H_0 . Roll out Variant B to production as the measured gain is statistically real and practically meaningful.

4. DELIVERABLES & REPOSITORY VERIFICATION

- Reproducible Dataset:** `ab_test_data.csv` (=200).
- GitHub Repository:** Live on <https://github.com/JeevanN-07/placemux.git>.